

Demonstration of Tests of Conversion of PM Dot Notation to Parentheses

Dennis J. Darland

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SECTION 0. VERIFICATION TESTS (of dot to paren dot icn)

For each proposition is given:

- 1: the PM notation with dots.
- 2: the notation with parentheses
- 3: the Polish (with Lukasiewicz symbols) notation

*2·06 $\vdash$:: $p \supset q . \supset : q \supset r . \supset . p \supset r$

*3·47 $\vdash$:: $p \supset r . q \supset s . \supset : p . q . \supset . r . s$

*4·22 $\vdash$:: $p \equiv q . q \equiv r . \supset . p \equiv r$

*4·41 $\vdash$:: $p . \vee . q . r : \equiv . p \vee q . p \vee r$

*4·43 $\vdash$:: $p . \equiv : p \vee q . p \vee \sim q$

*4·44 $\vdash$:: $p . \equiv : p . \vee . p . q$

*4·87 $\vdash$:: $p . q . \supset . r : \equiv : p . \supset . q \supset r : \equiv : q . \supset . p \supset r : \equiv : q . p . \supset . r$

*4·88 $\vdash$:: $p . q . \supset . r . \equiv : p . \supset . q \supset r : \equiv : q . \supset . p \supset r : \equiv : q . p . \supset . r$

*5·33 $\vdash$:: $p . q \supset r . \equiv : p : p . q . \supset . r$

From Landon D. C. Elkind's Paper in Russell: Vol. 43, no. 1, page 44

*431·441 $\vdash$:: $p \vee q . \equiv . r \supset s$

*431·442 $\vdash$:: $p . \vee . q \equiv r : \supset : s$

*431·443 $\vdash$:: $p \vee q . \equiv . r : \supset : s$

*431·444 $\vdash$:: $p : \vee : q \equiv r . \supset . s$

*431·445 $\vdash$:: $p : \vee : q . \equiv . r \supset s$

From same, page 54

*431·54 $\vdash$:: $p . q : r . s : \supset : p . s : r . q$

check longer prop name

Propositions involving quantifiers

$$*9\cdot2\vdash:(x).psix.\supset.psiy$$

$$*9\cdot21\vdash:(x).psix\supset phix.\supset:(x).psix.\supset.(x).phix$$

$$*9\cdot22\vdash:(x).psix\supset phix.\supset:(\mathfrak{A}x).psix.\supset.(\mathfrak{A}x).phix$$

$$*9\cdot31\vdash:(\mathfrak{A}x).phix.\vee.(\mathfrak{A}x).phix:\supset.(\mathfrak{A}x).phix$$

$$*9\cdot401\vdash:p:\vee:q.\vee.(\mathfrak{A}x).psix:\supset:q:\vee:p.\vee.(\mathfrak{A}x).psix$$

$$*10\cdot35\vdash:(\mathfrak{A}x).p.psix.\equiv:p:(\mathfrak{A}x).psix$$

$$*11\cdot2\vdash:(x,y).phi[x,y].\equiv.(y,x).phi[x,y]$$

One Step in proof of 11.55 I wanted example of 2 adjacent quantifiers -
hard to find.

$$*11\cdot551\vdash:(x):(\mathfrak{A}y).psix.phi[x,y].\equiv:psix:(\mathfrak{A}y).phi[x,y]$$

From same, page 46

$$*431\cdot46\vdash:(x).psix.phix.\supset.(x).psix$$

Other Tests

$$*99\cdot99\vdash:\sim(\mathfrak{A}x):\sim psix.\supset.(x).\sim psix$$